Assignment - 01

Name:- Rizwanul Islam

ID :- 21201129

Sec :- 12

Course :- CSE422

Answer to the Question no.01

a

$S_1 = S \rightarrow A \rightarrow D \rightarrow E \rightarrow G = 1+2+3+7 = 13$

$S_2 = S \rightarrow A \rightarrow B \rightarrow E \rightarrow G = 1+4+3+7 = 15$

$S_3 = S \rightarrow B \rightarrow E \rightarrow G = 3+3+7 = 13$

$S_4 = S \rightarrow B \rightarrow F \rightarrow G = 3+4+1 = 8$

$S_5 = S \rightarrow C \rightarrow F \rightarrow G = 5+1+1 = 7$

The admissibility is $h(n) \leq g(n)$

$\therefore h(s) = 4$

$A_1 = A \rightarrow D \rightarrow E \rightarrow G = 2+3+7 = 12$

$A_2 = A \rightarrow B \rightarrow E \rightarrow G = 4+3+7 = 14$

$A_3 = A \rightarrow B \rightarrow F \rightarrow G = 4+4+1 = 9$

$\therefore h(A) = 7$

$B_1 = B \rightarrow E \rightarrow G = 3+7 = 10$

$B_2 = B \rightarrow F \rightarrow G = 4+1 = 5$

$\therefore h(B) = 4$

$C = C \to F = 1+1 = 2$

$\therefore h(C) = 2$

$D = D \to E \to G = 3+7 = 10$

$\therefore h(D) = 9$

$E = E \to G = 7$

$\therefore h(E) = 6$

$F = F \to G = 1$

$h(F) = 1$

Therefore, the table will be –

states, n	h(n)
S	~~5~~ 4
A	7
B	4
C	2
D	9
E	6
F	1
G	0

b

Best First Search :-

$\boxed{S^4}$

S^4 $\boxed{A^7\ B^4\ C^2}$

C^2 $\boxed{A^7\ B^4\ F^1}$

F^1 $\boxed{A^7\ B^4\ G^0}$

G^0 $\boxed{A^7\ B^4}$

$\therefore\ S \rightarrow C \rightarrow F \rightarrow G = 7$

A* search

$\boxed{S^4}$

S^4 $\boxed{A^8\ B^7\ C^7}$

C^7 $\boxed{A^8\ B^7\ F^7}$

F^7 $\boxed{A^8\ B^7\ G^7}$

G^7 $\boxed{A^8\ B^7}$

$\therefore\ S \rightarrow C \rightarrow F \rightarrow G = 7$

C

Checking consistent.

$h(s) \leq c(s,n) + h(n)$

For S,

A

$4 \leq 1 + 8$

$5 \leq 9$

B

$5 \leq 3 + 4$

$5 \leq 7$

C

$5 \leq 5 + 2$

$5 \leq 7$

$\therefore$ h(s) is consistent.

For A,

B

$7 \leq 4 + 4$

$7 \leq 8$

D

$7 \leq 2 + 7$

$7 \leq 9$

$\therefore$ h(A) is consistent

For B,

E

$4 \leq 3 + 6$

$4 \leq 9$

F

$4 \leq 4 + 1$

$4 \leq 5$

$\therefore$ h(B) is consistent

For C,

F

$2 \leq 1 + 1$

$2 \leq 2$

h(C) is consistent

For D,

E

$9 \leq 3 + 6$

$9 \leq 9$

h(D) is consistent

For E,

G

$6 \leq 7 + 0$

$6 \leq 7$

$\therefore$ h(E) is consistent

For F,

G

$1 \leq 1 + 0$

$1 \leq 1$

$\therefore$ h(F) is consistent

$\therefore$ All the values are consistent

d

There will be no actual value for actual path as well as the heuristic value. As the goal node is not possible to reach the actual cost & the heuristic value will be ∞ (infinity).

Answer to the Question no. 02

a

For a better heuristic ~~value~~ function the value of a heuristic must be greater / equal to another heuristic value. Here $h_2(n)$ would be better, hence we can call it the dominance one. The corresponding value of $h_2(n)$ is greater or equal the values of $h_1(n)$. $h_2(A)$, $h_2(C)$, $h_3(D)$ are greater than their corresponding $h_1(n)$ values. $h_2(B)$, $h_2(E)$, $h_2(G)$ are equal.

b

A* Search

$\boxed{A^5}$

A^5 $\boxed{B^9 \; D^5}$

D^5 $\boxed{B^9 \; C^8 \; E^6 \; B^9}$

E^6 $\boxed{B^9 \; C^8 \; B^9 \; G^6}$

G^6 $\boxed{B^9 \; C^8 \; B^9}$

$\therefore A \rightarrow D \rightarrow E \rightarrow G = 6$

C

For A,

D	B
$5 \leq 1+4$	$5 \leq 3+6$
$\Rightarrow 5 \leq 5$	$\Rightarrow 5 \leq 9$

$\therefore$ h(A) is consistent

For B,

A	D	G
$6 \leq 3+5$	$6 \leq 2+4$	$6 \leq 9$
$\Rightarrow 6 \leq 8$	$\Rightarrow 6 \leq 6$	$\Rightarrow 6 \leq 9$

$\therefore$ h(B) is consistent

For C,

D
$5 \leq 2+4$
$\Rightarrow 5 \leq 6$

$\therefore$ h(C) in consistent

For D,

A	B	E
$4 \leq 1+5$	$4 \leq 2+6$	$4 \leq 4+1$
$\Rightarrow 4 \leq 6$	$\Rightarrow 4 \leq 8$	$\Rightarrow 4 \leq 5$

$\therefore$ h(D) in consistent

For E,

D	G
$1 \leq 4+4$	$1 \leq 1+0$
$\Rightarrow 1 \leq 8$	$\Rightarrow 1 \leq 1$

$\therefore$ h(E) in consistent

For G,

E	B
$0 \leq 1+1$	$0 \leq 9+6$
$\Rightarrow 0 \leq 2$	$\Rightarrow 0 \leq 15$

h(G) in consistent

$\therefore$ Choosen hearistic in consistent

Answer to the Question no. 3

a

The properties of Local Search:-

1. A single current state is being used & no path is needed
2. No need of back tracking
3. Time efficient
4. Finds a near good solution
5. The local search is needed for optimization problem

b

Some example of local search algorithm:-

1. Hill Climb Algorithm
2. Genetic Algorithm
3. Simulated Annealing

c

Drawback of hill climbing algorithm,

- Local Maxima:- It cannot detect the global highest point rather, it detects the highest point locally which is not the actual one.
- Platue:- The value does'nt increase / decrease as it is constant
- Ridges:- A special kind of maxima & has a area higher than its surrounding.

Remedies:-

- Untill the goal is found it restarts the search from a random location
- To eliminate problematic feature it reformulate the search space
- To avoid becoming trapped at local maxima it simulate annealingly

d

Local Maxima in 8 Puzzle :- In the puzzle a close to goal state might have fewer tiles which is misplaced than its neighbour which can seem optimal, it is not the goal. Random restart can solve the problem.

Platue :- The algorithm might enter a region where all the neighbouring nodes have some heuristic value for which leading to improvement to reach the goal is harder. The simulated annealing can solve it.

e

The key steps of simulated annealing are:-

(i) Initialitation

(ii) Neighborhood selection

(iii) Acceptance probability

(iv) Temperature decrease.

(v) Termination

f

ΔE = Energy diffenence

T = Tempenatune

$$P = e^{\Delta E/T}$$

~~If~~ when the value of T is high, this allow the algorithm to escape local optima by accepting local optima by accepting sub-optimal solution which is higher probability.

g

The relationship between tempenatune & the probabilistic →

- If the value of tempenatune is high, the value of $e^{\Delta E/T}$ will approach to 1 & the algorithm is more likely to accept worst solution

- If the value of tempenatune is low, the value of $e^{\Delta E/T}$ will appnoach to 0 & the algorithm only accept the better solution.